When *Um* Means Nothing – and Everything: Disfluency Judgments Across Minds and Modalities

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Disfluency refers to interruptions in the fluent speech stream, such as filled pauses (*um/uh*). Prior work argued that disfluency signals cognitive difficulties (Arnold et al., 2003) and lowers perceived speaker competence and trustworthiness (Charoenruk & Olson, 2018). These studies often used scripted, lab-recorded sentences that made disfluency salient: it is important to study effects of disfluency in spontaneous speech where it is ubiquitous (Zhang, 2020). Furthermore, disfluencies are increasingly encountered as text due to the rise of automatic transcription (e.g., in videos, or voice messages). As these verbatim transcripts are frequently fed into Large Language Models (LLMs), it is crucial to assess the interpretation and social evaluation of disfluency across modalities (audio vs. text) and "minds" (human vs. machine). One possibility is that, since reading and listening share a single, amodal linguistic mechanism (uniformity assumption; Gernsbacher et al., 1990; Gough & Tunmer, 1986) yielding identical parsing outcomes (Ferreira et al., 2002), modality might have limited effects on disfluency interpretation, at least for humans. Alternatively, the uniformity assumption may not extend to prosody: spoken disfluency is an ephemeral prosodic event that listeners rarely retain in conscious memory (Bard & Lickley, 1997), but text transcription turns it into a static, visual record. Research on "visual disfluency" (e.g., hard-to-read fonts) shows that static visual irregularities trigger negative affect (Song & Schwarz, 2008). Here, using naturalistic stimuli, we examine human evaluations of disfluency in speech (Exp.1) and text (Exp.2), as well as LLM judgments of disfluency (in text; Exp.3) to address these possibilities.

Exp.1 (Human-Audio): We extracted 10 spontaneous audio stimuli from corpora (5 female speakers, 5 casual speech), each 15-20 seconds long (M = 50.80 words), containing 4 filled pauses (*FP*) in accordance with naturalistic rates (Zhang, 2020) (e.g., Table 1). We next digitally removed the filled pauses to create a *No-FP* version. A 2x2 design crossed Filled Pause (*FP* vs. *No-FP*) with Speaker Expertise (Expert vs. Novice), which was described by text prior to presenting each auditory stimulus (Table 2). 100 undergraduates listened to the audio stimuli and rated speaker competence, warmth, and trustworthiness on a 1-7 Likert scale (18 items total; McCroskey & Teven, 1999). Results were analyzed using Structural Equation Modeling (SEM). Filled Pause had no main effects or interactions (all $ps > .05$; Fig.1). Experts were judged higher than novices in competence ($p = .02$) but not in warmth ($p = .17$) or trustworthiness ($p = .23$). Listeners do not penalize disfluency in spontaneous speech in terms of global speaker judgments.

Exp.2 (Human-Text): 100 new participants read transcripts of the audio stimuli from Exp.1. Results (Fig.2) revealed a dramatic reversal: filled pauses lowered perceived speaker competence ($p < .001$), warmth ($p = .004$), and trustworthiness ($p < .001$). Expertise had no effect on any construct, nor did it interact with Filled Pause ($ps > .05$). Contrary to the uniformity assumption, readers treated textual disfluency differently from spoken disfluency – perhaps as a visual disruption triggering a broad negative bias and preempting deeper processing of Expertise.

Exp.3 (LLM-Text): We replicated Exp.2 with GPT-4o (100 simulated participants, T = .5). Results (Fig.3) showed that filled pauses lowered perceived speaker competence ($p = .002$) but had no effect on warmth or trustworthiness ($ps > .05$). The LLM judged experts as higher in competence ($p < .001$) without other main or interaction effects ($ps > .05$). In contrast to humans, the LLM did not penalize disfluency in text across the board (and preserved expertise effects). It penalized only speaker competence, presumably driven by training data where disfluency statistically co-occurs with markers of low confidence (cf. Brown et al., 2020).

Discussion. Disfluency functions differently across modalities and minds. In human comprehenders, social experience affords tolerance to spoken – but not written – disfluency; in LLMs, disfluencies are not interpreted the way humans interpret textual disfluencies but in ways that approximate some degree of experience with spoken language.

Table 1. Sample stimulus (*FP* vs. *No-FP*) that adults listened to (Exp.1) or read (Exps.2 and 3).

(Um), when we have to prove something like specific intent, **(um)**, or we have to prove identity, because he's disputing identity, at least one of the cases. **(Um)**, then, we are allowed to bring in prior similar conduct, **(um)**, where he acted in an identical way, or where the victims were in a similar situation. (55 words)

Table 2. Sample text for different levels of speaker expertise (all Exps.).

Expert	Rebecca is an experienced Philadelphia attorney. A client is currently involved in a lawsuit. Rebecca is offering an opinion on the case.
Novice	Rebecca is working as an intern in a law firm. A client is currently involved in a lawsuit. Rebecca is offering an opinion on the case.

Fig.1. Results of Exp.1.

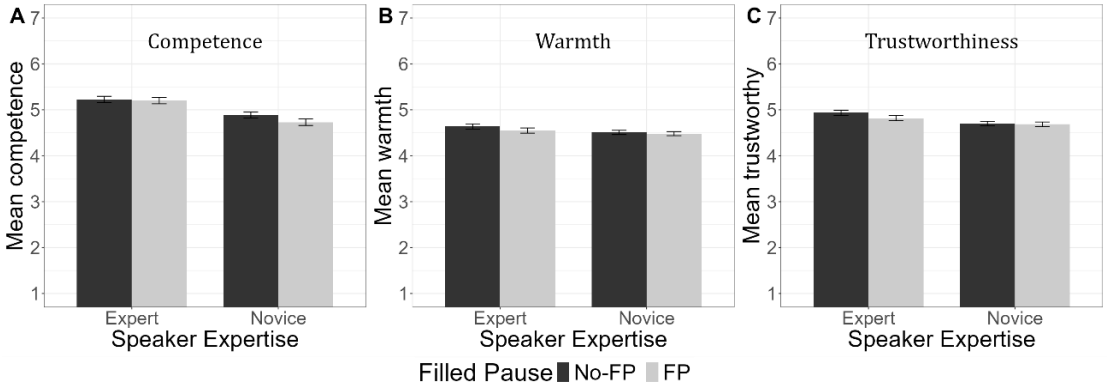

Fig.2. Results of Exp.2.

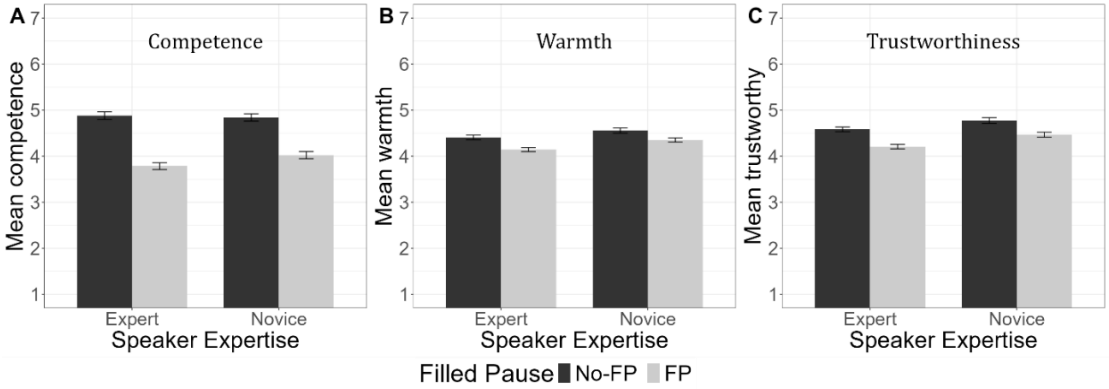
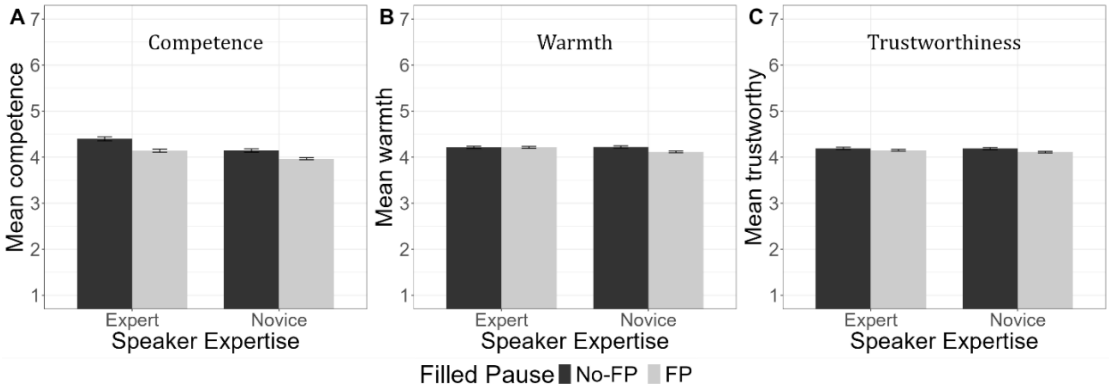
Fig.3. Results of Exp.3.